Physical Chemistry CUME, April. 24, 2015

Basic Concepts in Thermodynamics

1. A system in equilibrium at temperature T is expected to satisfy the Boltzmann distribution $p_i = \frac{1}{q} e^{-\beta E_i}$, where p_i is the probability of microstate i, E_i is the energy of microstate i and q is the partition function, which can be considered as a normalization factor.

A. What is β in the Boltzmann distribution?

B. Consider a two state problem with ground state energy E_0 and excited state energy E_1. What is the partition function of the two state system?

C. Assume E_0 is 0, E_1 is 10^{-20} J, what is the energy of the system at 0K, 1000K, and T=infinity. (Hint: $R = k_B \cdot N_A$ in case you do not remember the numerical value for k_B)

D. Assuming a laser causes a population inversion, where the ground state population is 10% and the excited state population is 90%, what is the effective temperature of the two state system mentioned in C?

2. In an isolated system, which is frequently referred to as the microcanonical ensemble, the entropy of the system is $S = k_B \ln(W)$ where k_B is the Boltzmann's constant. Two different gas molecules A and B are separated into two chambers with identical volume *V*. Assuming there are *N* molecules of A, and *N* molecules of B. Assume a valve is opened that allowed A and B to mix. What is the entropy of mixing?

Assume N is large, and the Sterling approximation is $\ln(N!) = N\ln(N)$